



Girls in Security Operations (GiSOC) Bootcamp

Cyberblue Log Analysis Using Splunk

By

Analyst: ALI CHINASA JULIET

Date: Novemeber 14th, 2025

Submitted as part of the Final Capstone Project for the Girls in Security Operations Bootcamp.

Table of Contents

1. Executive Summary

2. Log Overview

3. Key Findings

3.1 Multiple Failed Login Attempts

3.2 Pre-Authentication Disconnects

3.3 Suspicious IP Activity

3.4 All Source IPs

3.5 Authentication Trends

4. Incident Investigation Summary

4.1 OTX Investigation

4.2 Recommended Actions

5. Dashboard Summary

6. Alert Configuration

7. Field Extraction

8. Investigation Summary

8.1 Observations

8.2 Potential attacks

9. Recommendations

9.1 Immediate Actions

9.2 Near Term

9.3 Long Term

10. Summary

1. Executive Summary

This investigation analyzed SSH logs using Splunk Enterprise to detect suspicious authentication activity and disconnect patterns. Dashboards, alerts, and field extractions were created for continuous monitoring.

Key findings:

- Repeated SSH pre-authentication failures.
- Multiple failed logins attempt from external IPs.
- IPs **183.62.140.253** and **187.141.143.180** flagged for scanning and brute-force activity
- Further investigation on OTX confirmed both IPs as active threats.

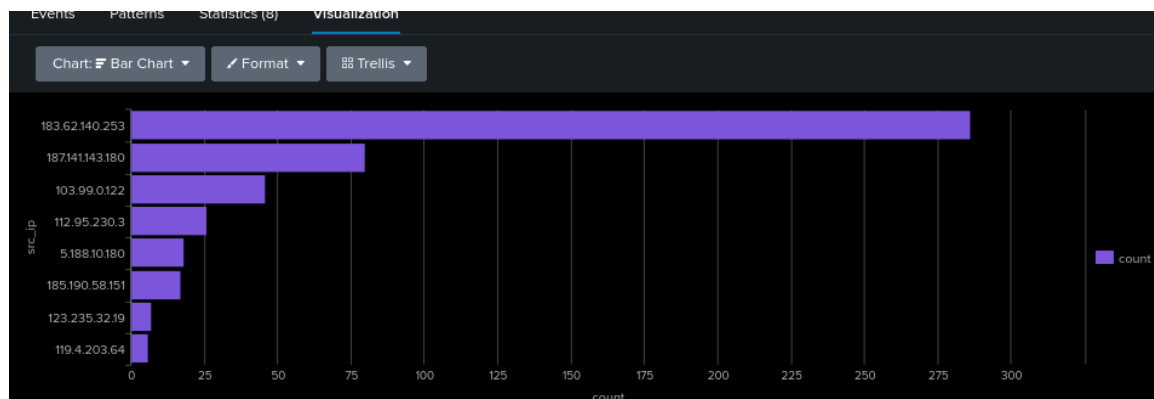
2. Log Overview

3. Key Findings

- Logs included SSH authentication events with successful, failed, and pre-auth attempts.
- Most suspicious activity originated from external IPs linked to internet-wide scanning.
- Data was indexed in Splunk's main index and timestamps were parsed for time-based analysis.

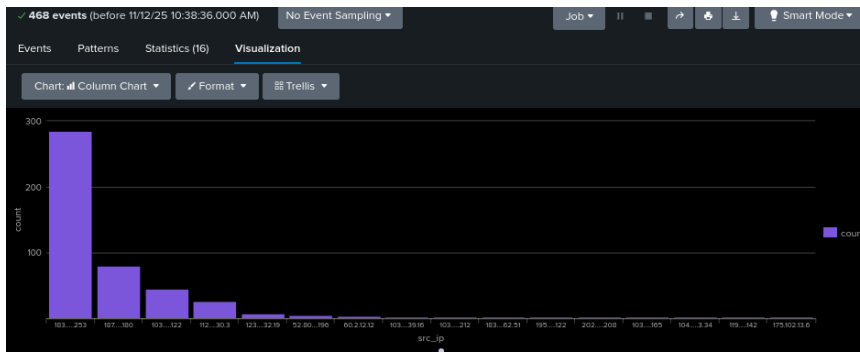
3.1 Multiple Failed Login Attempts

- Several IPs attempted multiple passwords in short intervals.
- Targets included root and generic accounts.
- Activity aligns with automated tools such as Hydra or generic SSH scanners.



3.2 Pre-Authentication Disconnects

- Consistent messages like "Received disconnect from 112.95.230.3: 11: Bye Bye [preauth]".
- Indicates automated scanning and fingerprinting before brute-force attacks.



New Search

index="main" "Received disconnect" | rex field=_raw "(?<src_ip>\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3})" | search src_ip
="183.99.0.122"

45 events (before 11/12/25 10:44:37.000 AM) No Event Sampling

Events (45) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection X Deselect 1 minute per column

Format Show: 20 Per Page View: List

< Prev 1 2 3 Next >

i	Time	Event
>	11/12/25 11:04:41.000 AM	1989,Dec,10,11:04:41,LabSZ,25534,error: Received disconnect from 103.99.0.122: 14: No more user authentication methods available. [preauth],E7,error: Received disconnect from <*>; <*>; No more user authentication methods available. [preauth] host = kali ; source = CyberBlue Logs.csv ; sourcetype = csv
>	11/12/25 11:04:37.000 AM	1977,Dec,10,11:04:37,LabSZ,25527,error: Received disconnect from 103.99.0.122: 14: No more user authentication methods available. [preauth],E7,error: Received disconnect from <*>; <*>; No more user authentication methods available. [preauth] host = kali ; source = CyberBlue Logs.csv ; sourcetype = csv
>	11/12/25 11:04:33.000 AM	1968,Dec,10,11:04:33,LabSZ,25521,error: Received disconnect from 103.99.0.122: 14: No more user authentication methods available. [preauth],E7,error: Received disconnect from <*>; <*>; No more user authentication methods available. [preauth] host = kali ; source = CyberBlue Logs.csv ; sourcetype = csv
>	11/12/25 11:04:28.000 AM	1956,Dec,10,11:04:28,LabSZ,25513,error: Received disconnect from 103.99.0.122: 14: No more user authentication methods available. [preauth],E7,error: Received disconnect from <*>; <*>; No more user authentication methods available. [preauth] host = kali ; source = CyberBlue Logs.csv ; sourcetype = csv
>	11/12/25 11:04:23.000 AM	1944,Dec,10,11:04:23,LabSZ,25505,error: Received disconnect from 103.99.0.122: 14: No more user authentication methods available. [preauth],E7,error: Received disconnect from <*>; <*>; No more user authentication methods available. [preauth]

SELECTED FIELDS
a host 1
a source 1
a sourcetype 1

INTERESTING FIELDS
a Date 1
date_hour 2
date_mday 1
date_minute 4
a date_month 1
date_second 35
a date_wday 1
date_year 1
a date_zone 1
a Day 1
a EventId 1
a EventTemplate 1
a extracted_Host 1
a index 1
linecount 1
_source 15

3.3 Suspicious IP Activity

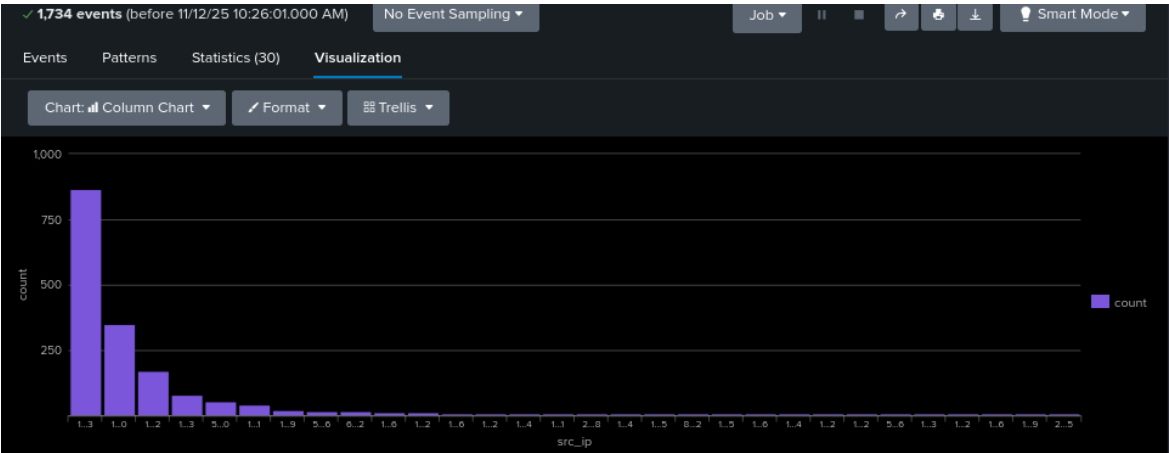
- 183.62.140.253: repeated failed logins, disconnects, flagged by VirusTotal.
- 187.141.143.180: repeated authentication failures, community reports of abusive behavior

Statistics (30)	
src_ip	count
183.62.140.253	867
187.141.143.180	349
103.99.0.122	172
112.95.230.3	80
5.188.10.180	53
185.190.58.151	43
123.235.32.19	22
52.80.34.196	15
60.2.12.12	15
103.207.39.16	12
103.207.39.212	12
173.234.31.186	10
195.154.37.122	10
119.4.203.64	9
183.136.162.51	8
202.100.179.208	8
104.192.3.34	7
103.207.39.165	5
88.147.143.242	5
186.5.5.105	4

Both IPs were further analyzed on OTX, confirming history of brute-force activity and association with threat pulses.

3.4 All Source IPs

High concentration of attempts from cloud-hosted international IPs



src_ip	count
183.62.140.253	867
187.141.143.180	349
103.99.0.122	172
112.95.230.3	80
5.188.10.180	53
185.190.58.151	43
123.235.32.19	22
52.80.34.196	15
60.2.12.12	15
103.207.39.16	12
103.207.39.212	12
173.234.31.186	10
195.154.37.122	10
119.4.203.64	9
183.136.162.51	8
202.100.179.208	8
104.192.3.34	7
103.207.39.165	5
88.147.143.242	5
186.5.5.185	4

3.5 Authentication Trends

Dashboards show continuous automated probing targeting SSH endpoints.

4. Incident Investigation Summary

During routine log analysis, the IPs **187.141.143.180** and **183.62.140.253** were identified for suspicious activity matching brute-force patterns.

4.1 OTX Investigation

- 183.62.140.253: Registered in Lianjiang, China, linked to uninet s.a. de c.v.
- 187.141.143.180: Registered in Mulege, Mexico, linked to uninet s.a. de c.v.

Both IPs are associated with brute-force attacks, honeypot activity, and related attack types.

IPV4

187.141.143.180

Add to Pulse

Analysis Overview

Reverse DNS

customer-187-141-143-180-sta.uninet-ide.com.mx

Location

Mexico, Mulege

ASN

AS8151 uninet s.a. de c.v.

DNS Resolutions

1 Domain

Related Pulses

OTX User-Created Pulses (34)

Related Tags

8 Related Tags
 honeypot, kfsensor, rdp, ssh, SSH, Honeypot, bruteforce, Bruteforce

Indicator Facts

Historical OTX telemetry
 1 domains resolved in all time
 1 top-level domains

External Resources

Whois, VirusTotal

OTX shows 34 threat pulses and tags including RDP, SSH, and brute force.

Historical data confirms these IPs represent a genuine threat to our environment

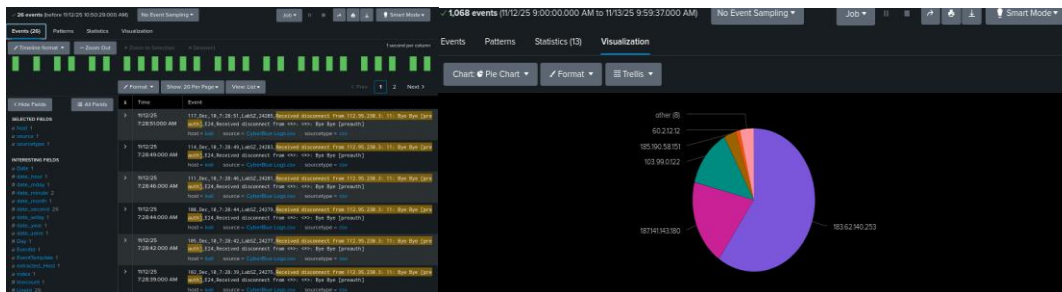
4.2 Recommended Actions

- Block both IPs at the network perimeter.
- Review all logs for possible connections to these IPs.
- Increase monitoring for related activity.

5. Dashboard Summary

Panels created:

- Top Source IPs by Failed Attempts
- Authentication Failures Over Time
- Received Disconnect Events
- Pre-auth Activity Overview
- Raw Event Table
- VirusTotal and OTX-Enriched Indicators



These allow rapid identification of spikes and isolation of malicious IPs.

6. Alert Configuration

- Search: "Received disconnect from 112.95.230.3: 11: Bye Bye [preauth]"
- Runs daily at 8 AM
- Trigger threshold: more than 2 events in 24 hours
- Email notifications restricted to approved domain
- 24-hour alert suppression to avoid duplicates

8.2 Potential attacks

- SSH password guessing
- SSH fingerprinting
- Enumeration of valid accounts

No successful compromise detected. The host is actively targeted.

9. Recommendations

9.1 Immediate Actions

- Block malicious IPs at the firewall
- Enforce SSH key authentication
- Disable root password login
- Enable fail2ban or equivalent protection

9.2 Near Term

- Ingest firewall and IDS logs into Splunk
- Add correlation searches for repeated failed logins, pre-auth disconnects, and logins outside business hours

9.3 Long Term

- Implement SSH MFA where possible
- Rotate credentials and audit privileged accounts
- Conduct vulnerability scans and patch reviews
- Continue threat intelligence enrichment

10. Summary

Splunk analysis confirms continuous external scanning and brute-force attempts. OTX and VirusTotal enrichment validate the top IPs as active threats. Dashboards, alerts, and field extractions now provide ongoing visibility and early warning for similar events.